



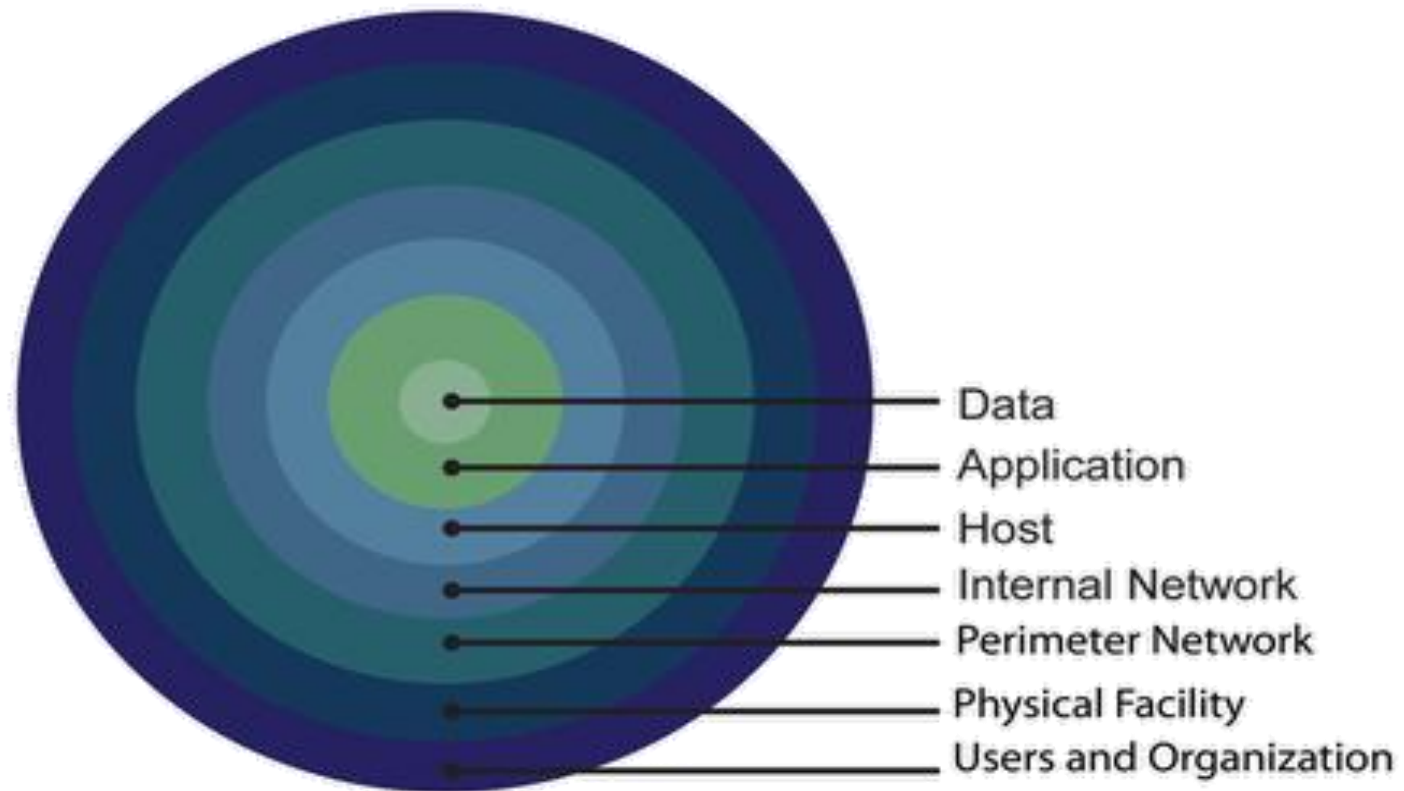
תשתיות וארכיטקטורה להגנת הסייבר א'

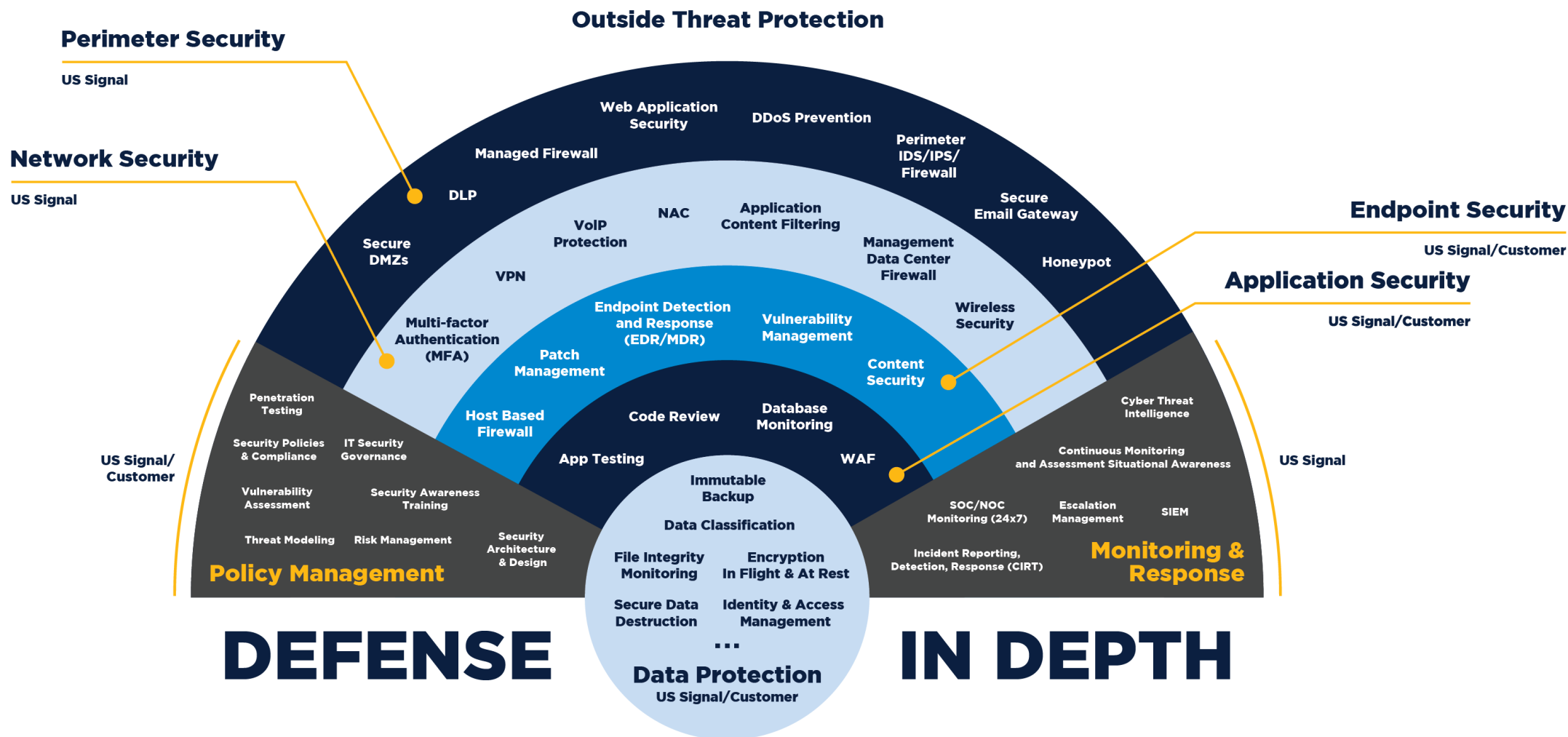
נושא 4

עקרונות הגנה

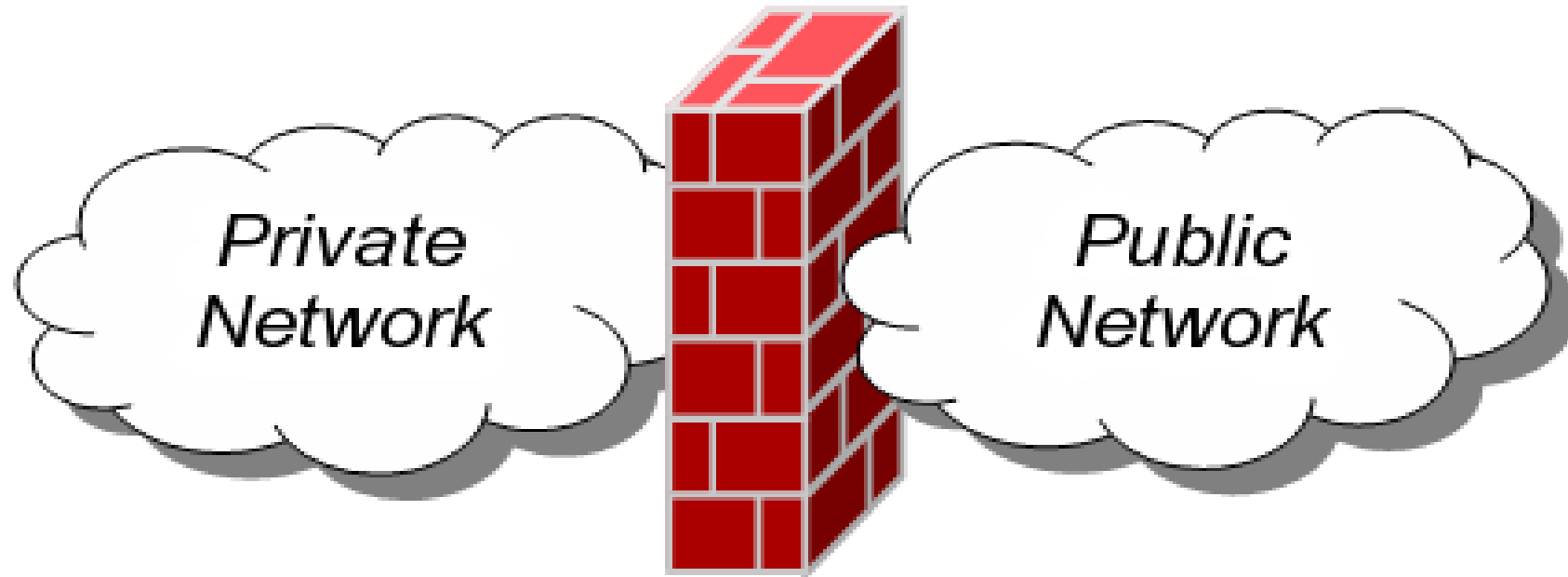
Defense in Depth, FW

Defense in depth



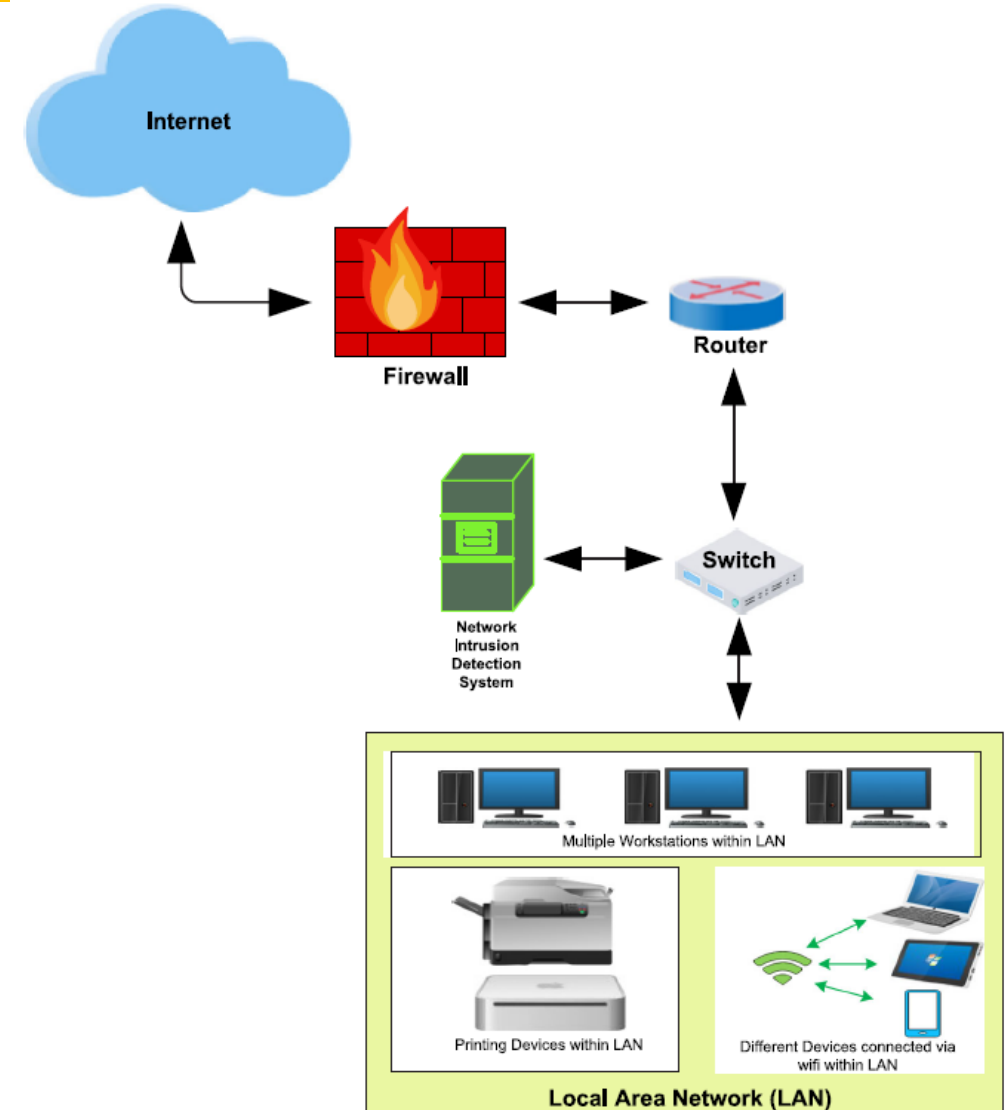
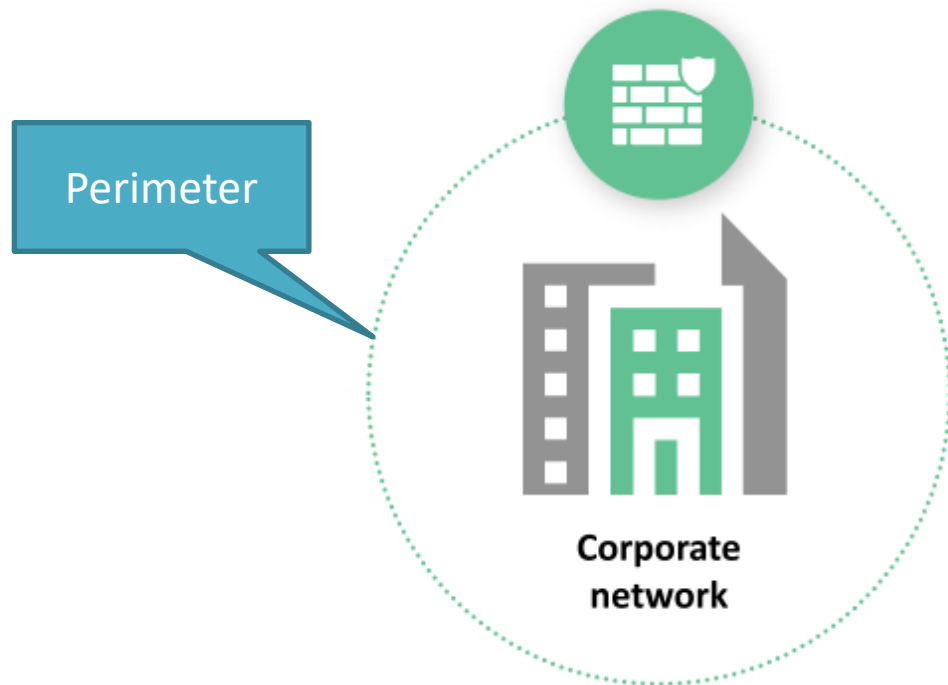


Traditional Model



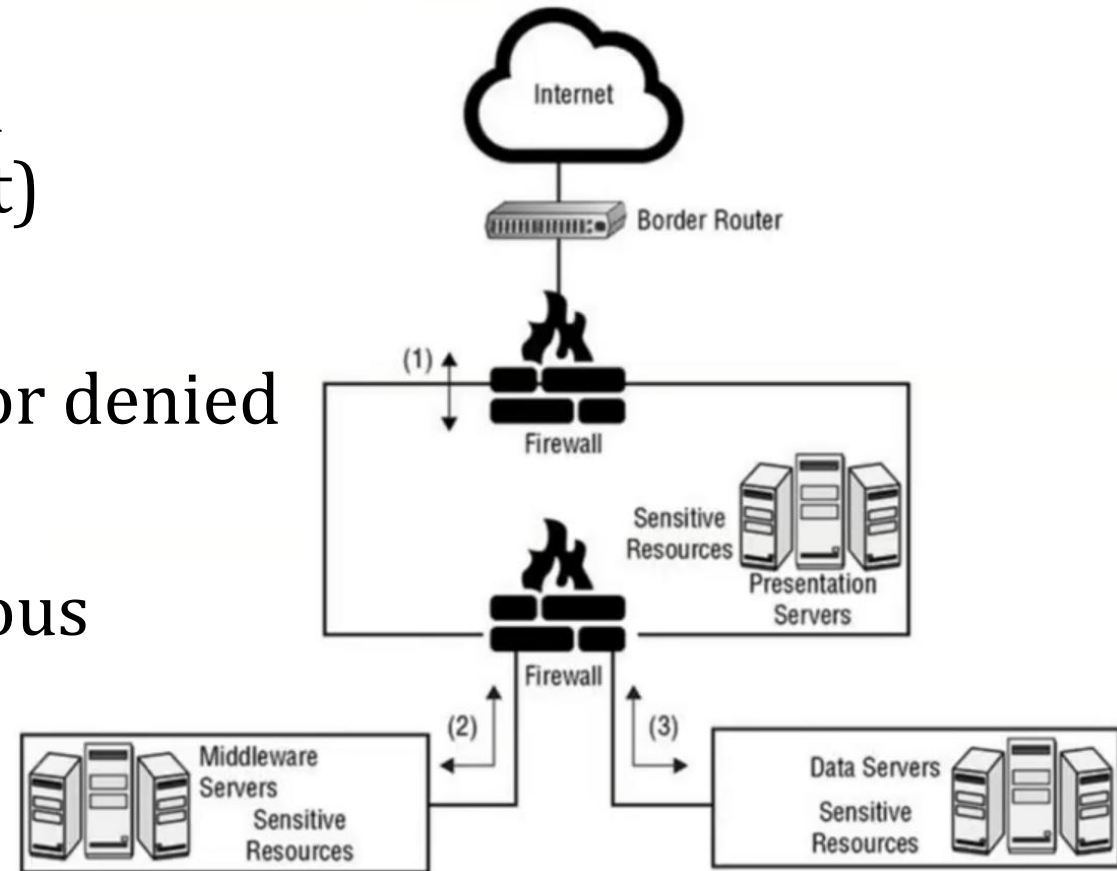
Traditional Model

Users, Devices, Apps, Data,
protected behind Perimeter



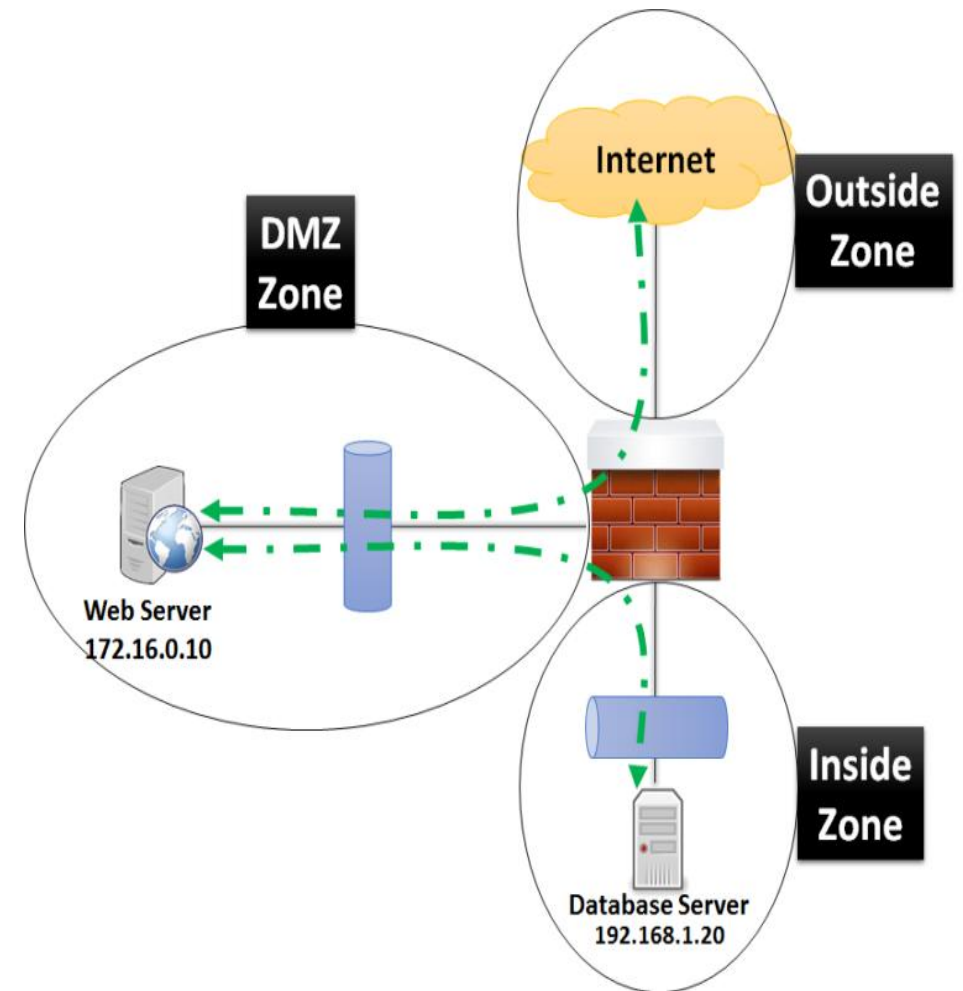
Elements of Perimeter Defense

- Border Routers:
 - the last router you control before an untrusted network (such as Internet)
- Firewalls:
 - decide what traffic is to be allowed or denied
- Intrusion detection system
 - an alarm system that detects malicious events and alerts



Elements of Perimeter Defense

- Intrusion Prevention Systems
 - provide automatic defense without administrators' involvements
- Virtual Private Networks
 - protected network session formed across an unprotected channel such as Internet
 - hosts connected through VPN are part of borders
- De-militarized zones (DMZ)
 - small network providing public services (not protected by firewall)





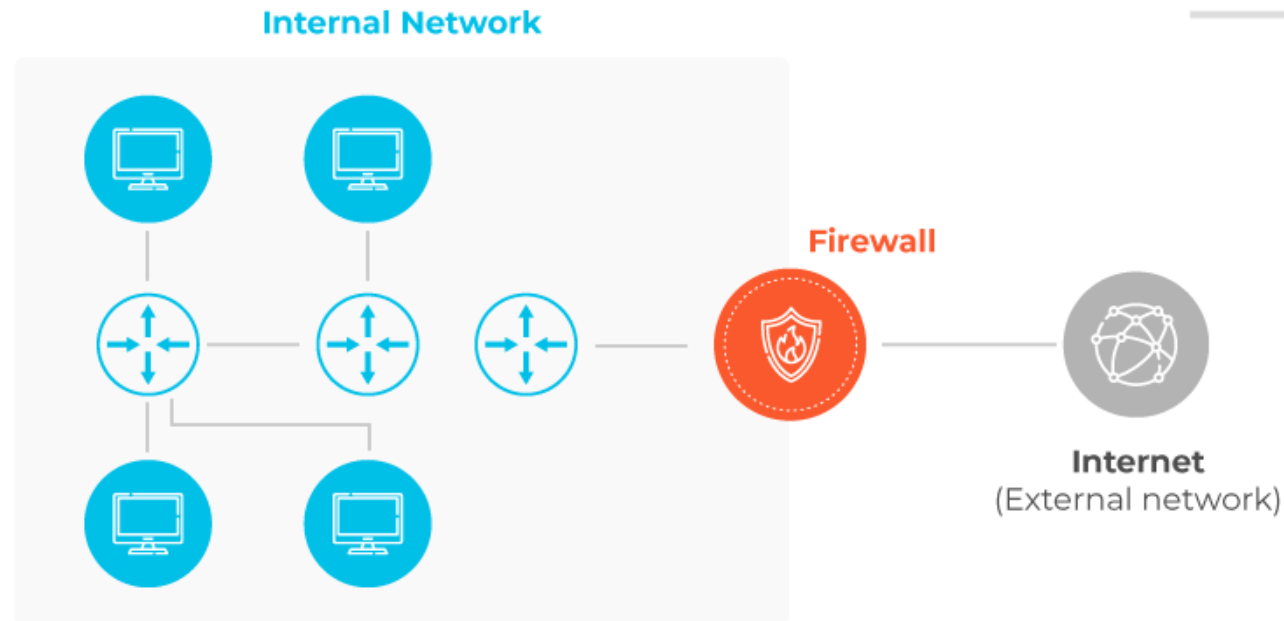
אבטחת מידע ממוחשב

Firewalls

Firewalls - layer 3

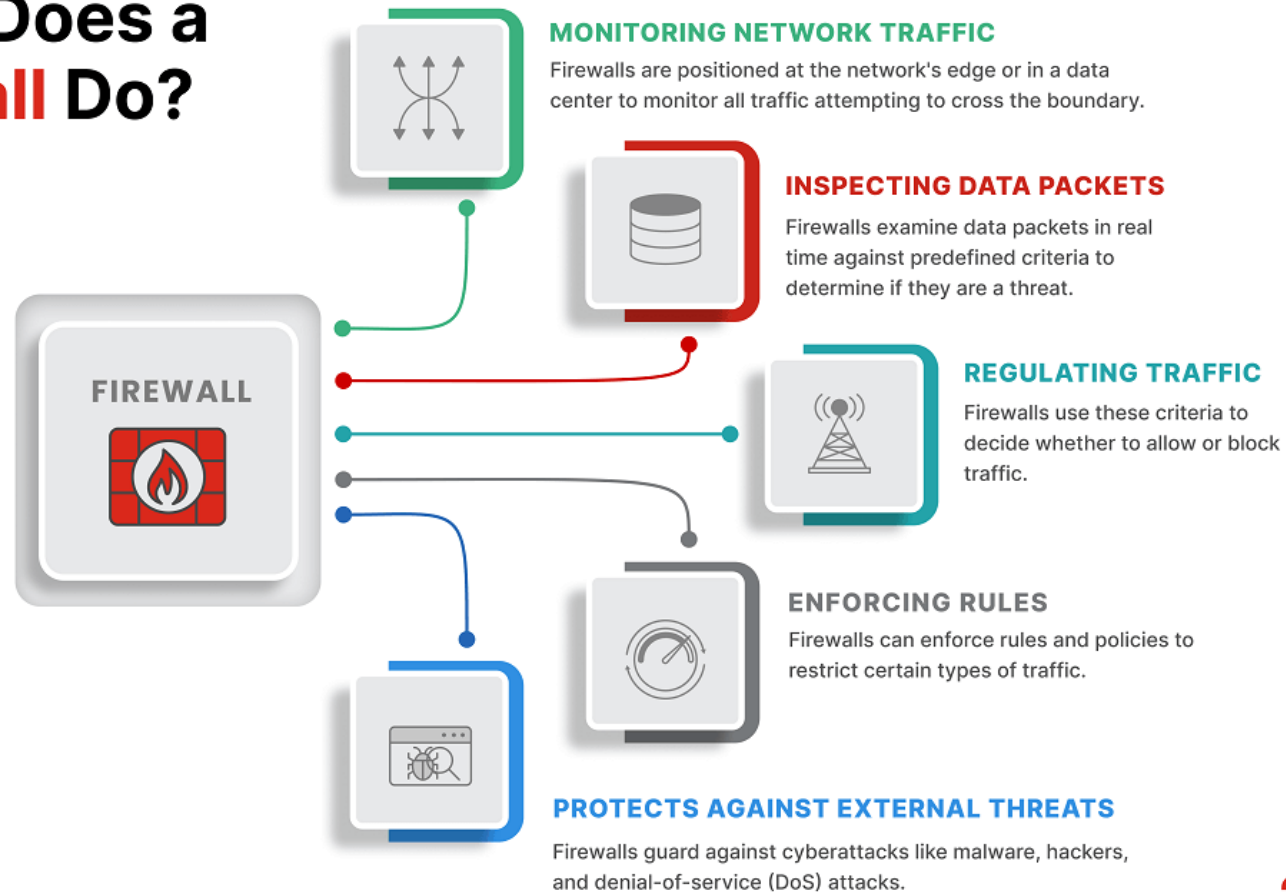
firewall

isolates organization's internal network from Untrusted Network (Internet/External Network), allowing some packets to pass, blocking others.

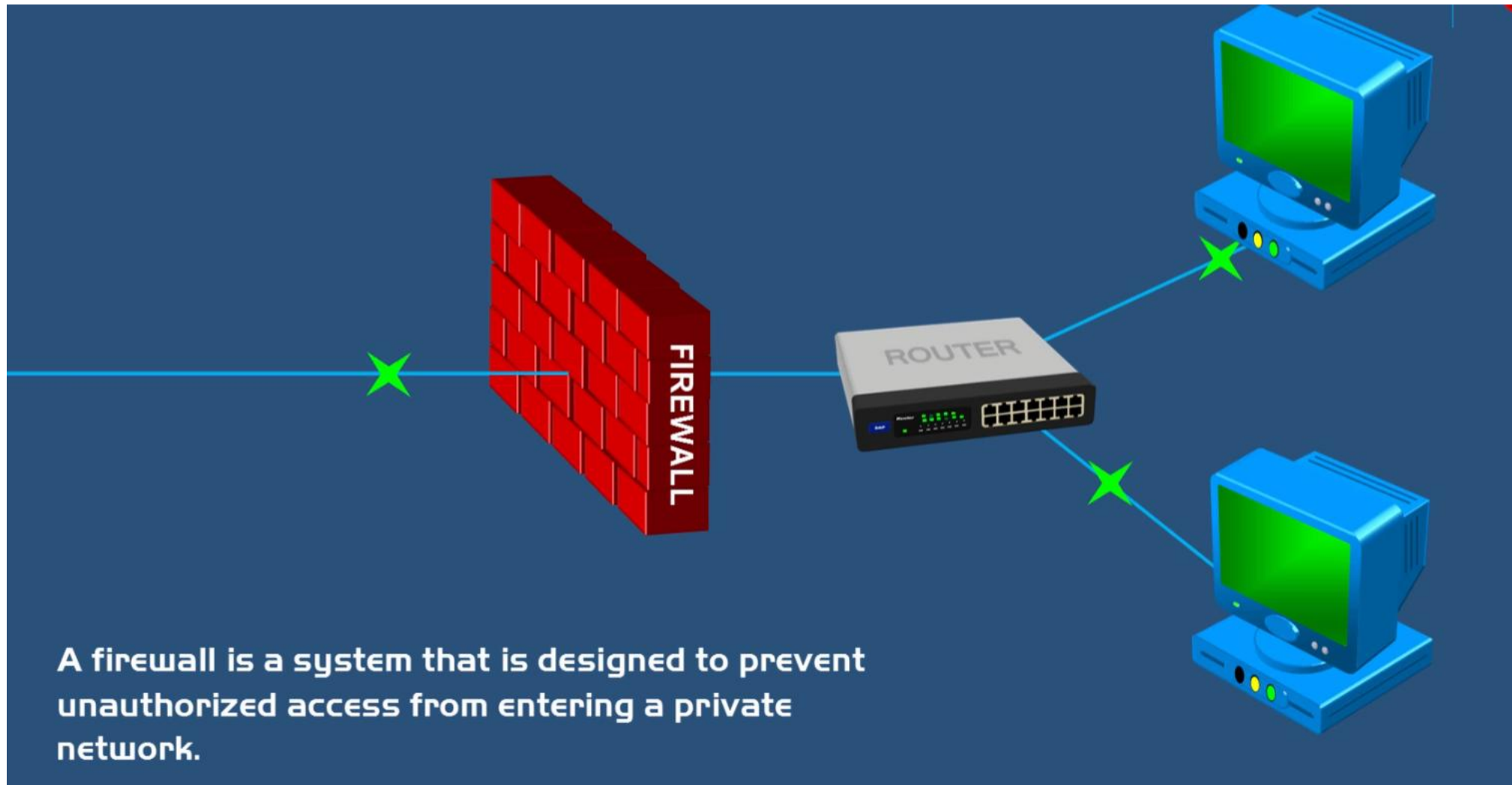


Firewalls - layer 3

What Does a Firewall Do?

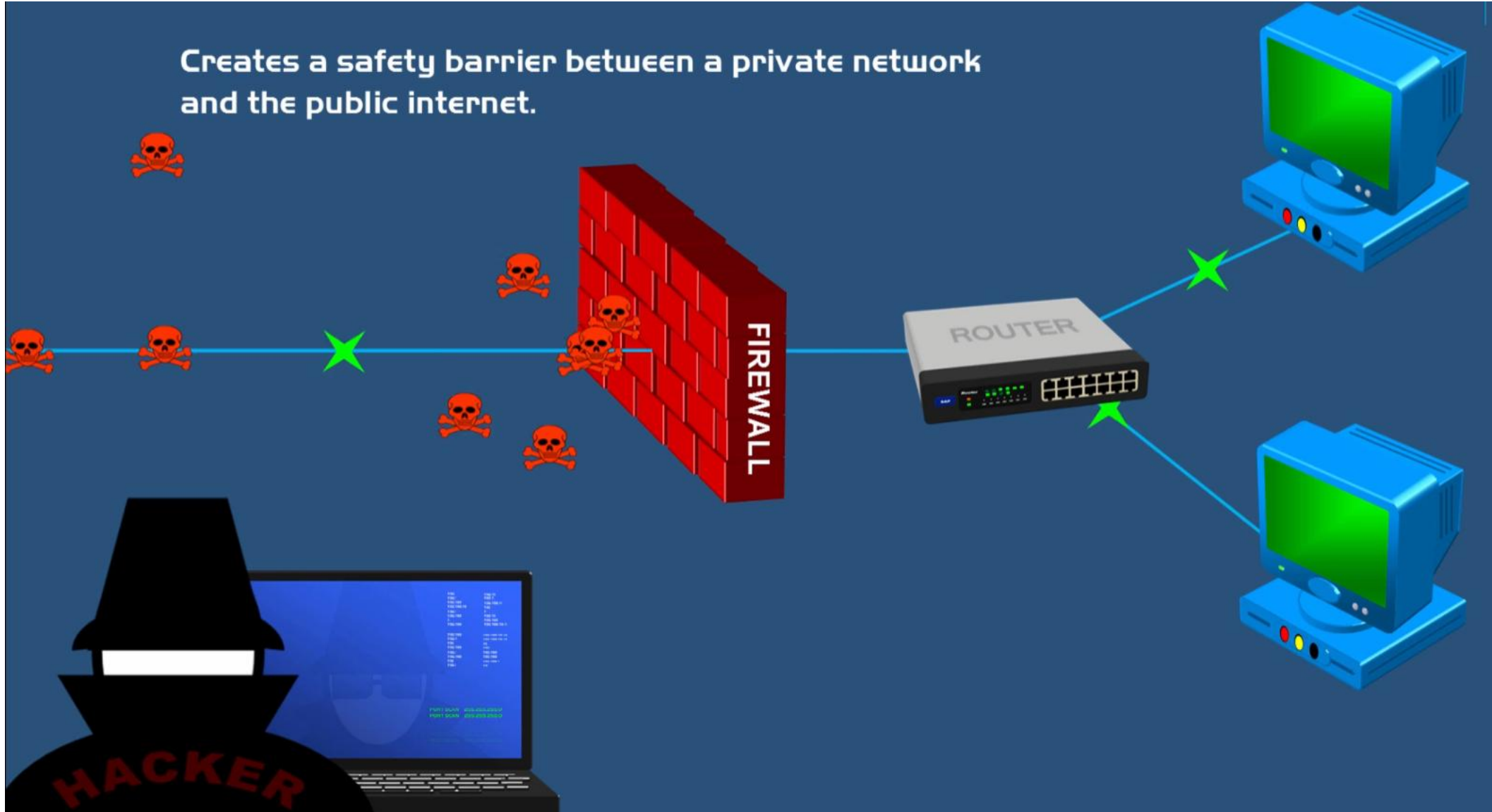


Firewalls - layer 3



Firewalls - layer 3

Creates a safety barrier between a private network and the public internet.

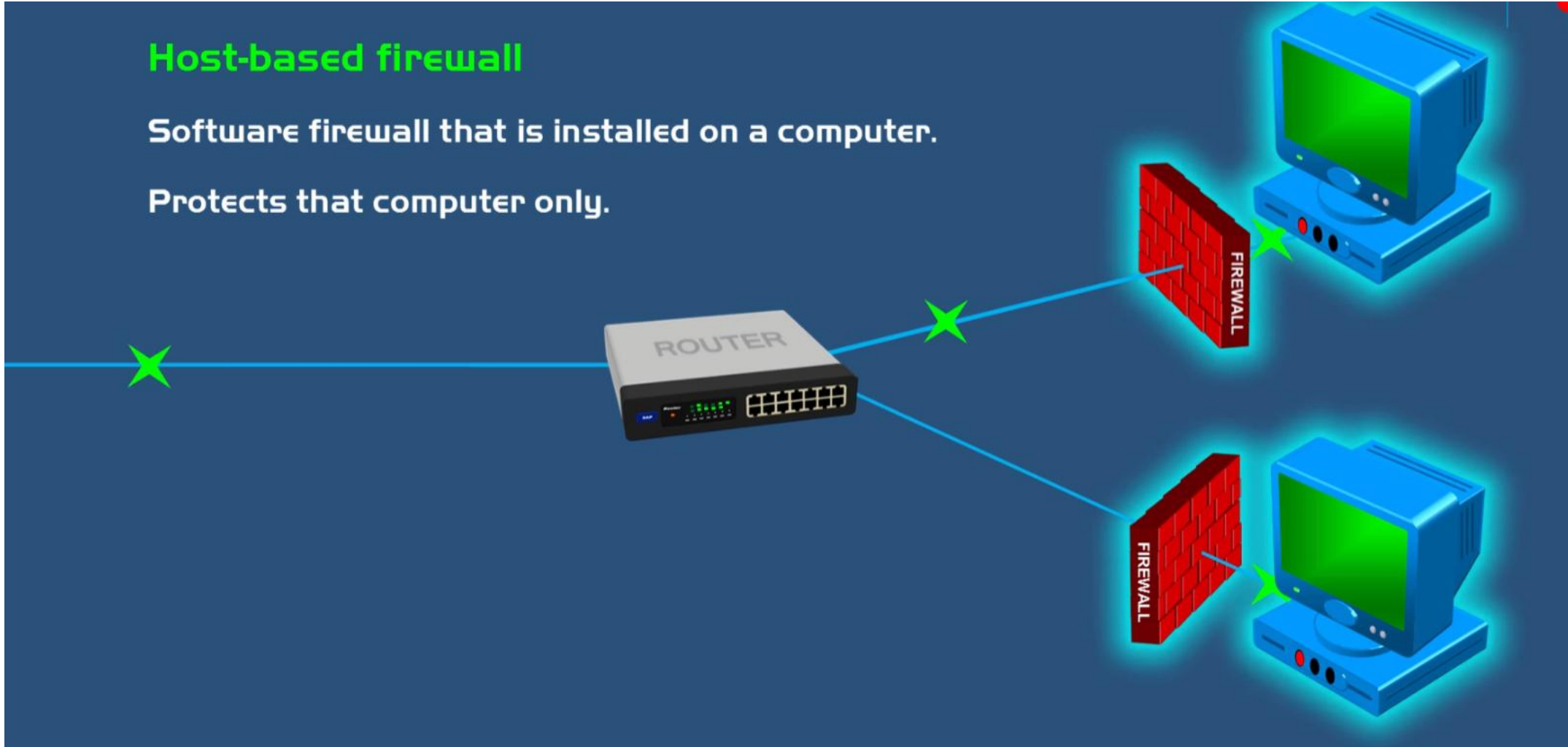


Host Based Firewalls

Host-based firewall

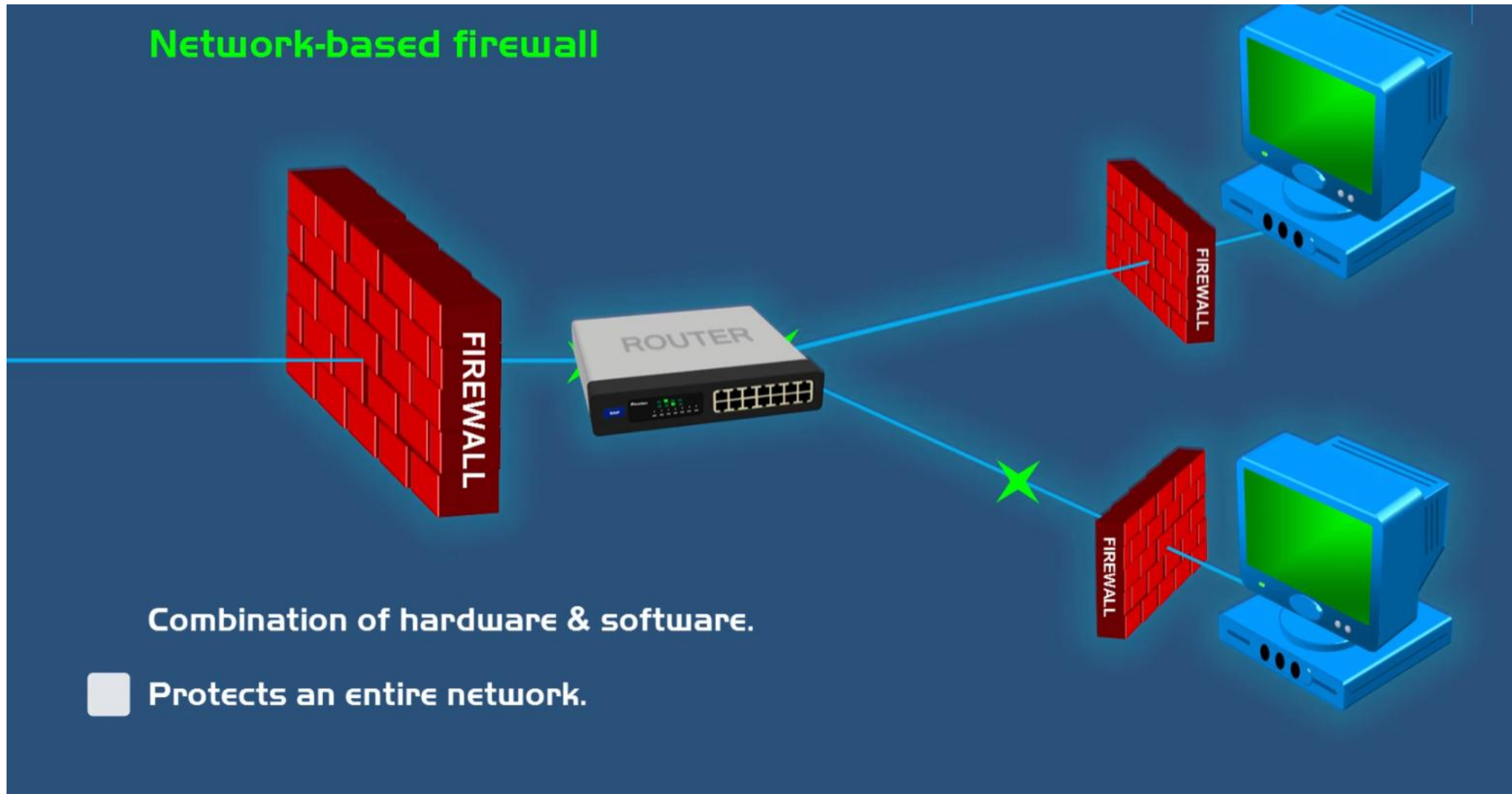
Software firewall that is installed on a computer.

Protects that computer only.



Firewalls - layer 3

Network-based firewall



Firewall Types



**Stand-alone
firewall**

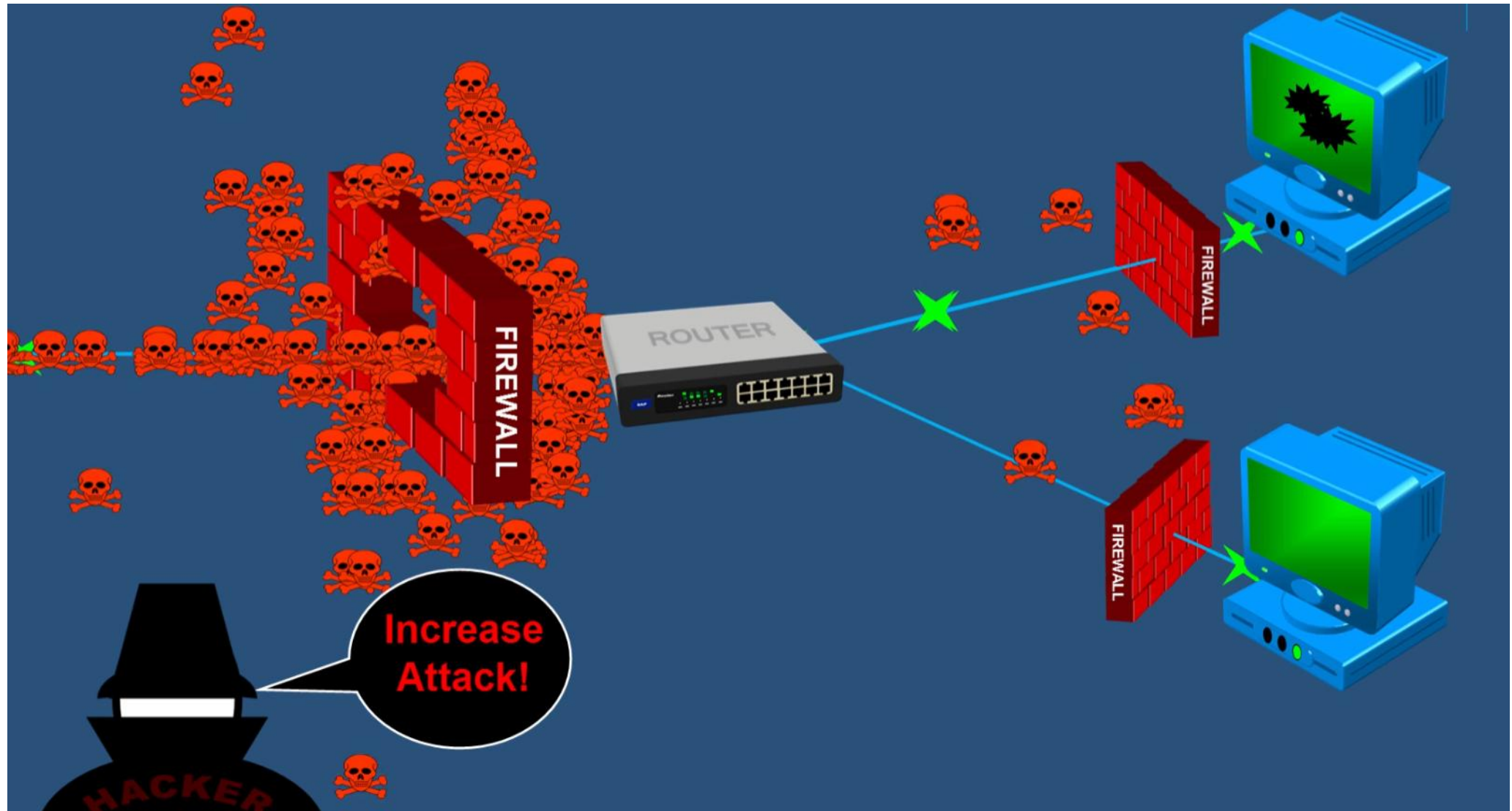


**Routers have a
built-in firewall**



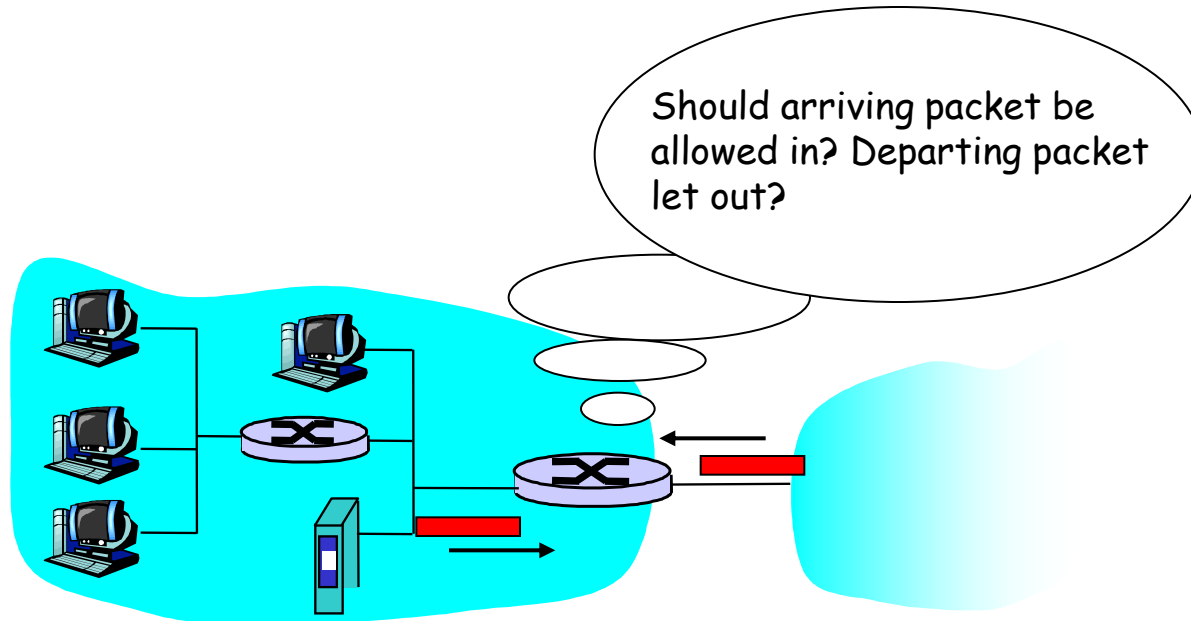
Cloud firewall

Firewall Breaches



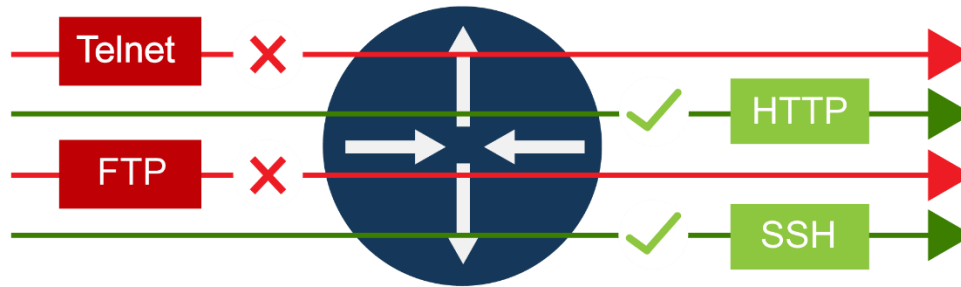
Packet Filtering FW

- internal network connected to Internet via **firewall**
- **filters packet-by-packet network** traffic and payload, decision to forward/drop packet based on:
 - source IP address
 - destination IP address
 - Port used
 - protocols



Network access control lists

Access control lists (ACLs) are a list attached to a resource, giving permissions or rules about precisely who can access it.



ACL rules

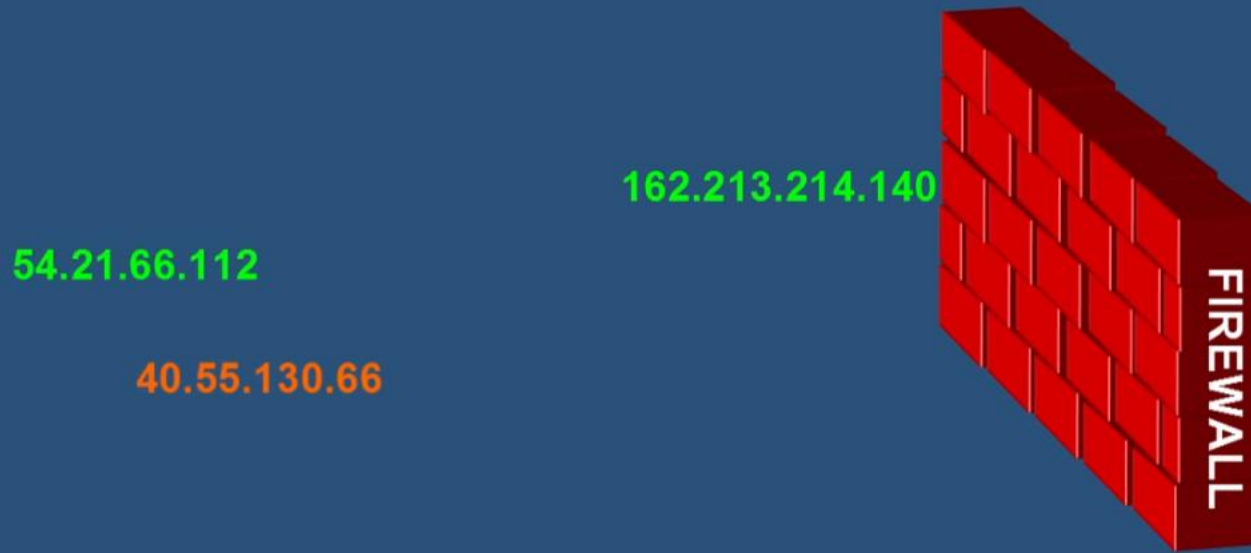
- Implicit Deny
 - Access is denied unless a rule explicitly allows it. An ACL containing only explicit allowances is often called a **whitelist** or allow list.
- Implicit Allow
 - Access is allowed unless a rule explicitly denies it. An ACL containing only explicit denials is called a **blacklist** or block list.

Packet Filtering FW Rules

ACCESS CONTROL LIST

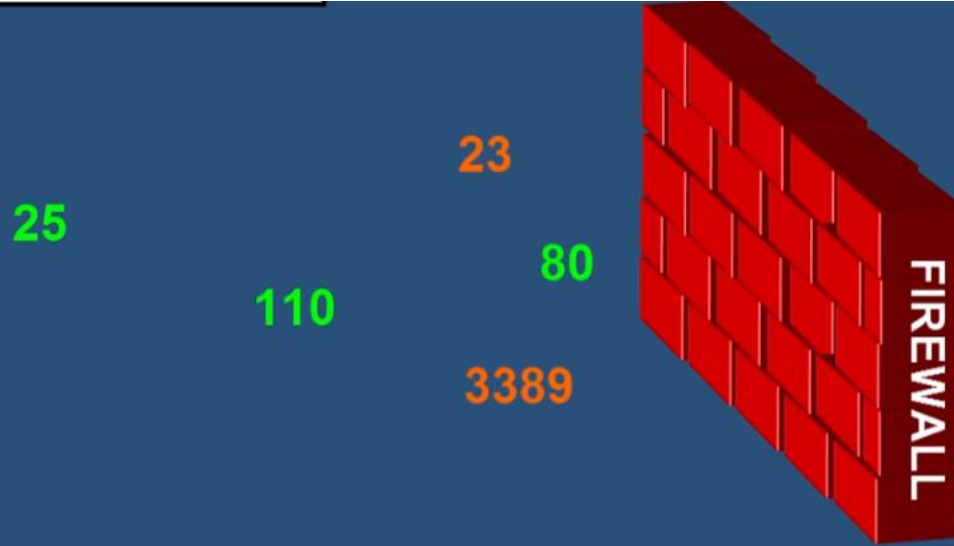
Permission	IP Address	Protocol	Destination	Port
ALLOW	162.213.214.14	TCP	10.10.10.2	80
ALLOW	54.21.66.112	TCP	ANY	80
DENY	192.168.1.1	TCP	ANY	80
ALLOW	65.252.1.2	TCP	ANY	80
DENY	ANY	TCP	ANY	80
ALLOW	ANY	TCP	ANY	80
DENY	ANY	UDP	10.10.10.1	23
DENY	255.255.255.0	TCP	ANY	25
ALLOW	10.10.0.1	TCP	ANY	110

Packet Filtering FW Rules



Permission	IP Address	Protocol	Destination	Port
ALLOW	162.213.214.140	TCP	ANY	80
ALLOW	54.21.66.112	TCP	ANY	80
DENY	40.55.130.66	TCP	ANY	80

Packet Filtering FW Rules

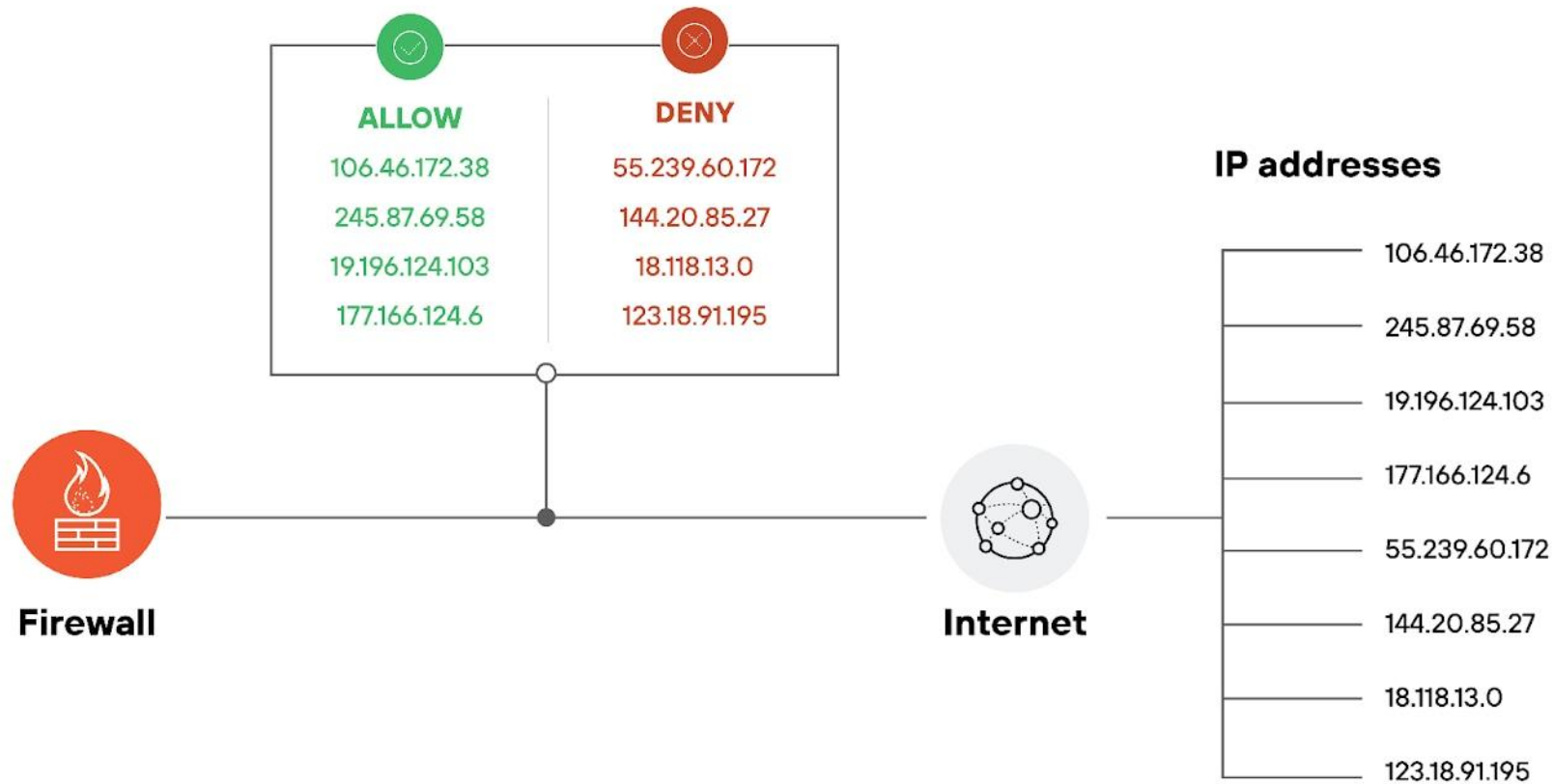


Permission	IP Address	Protocol	Destination	Port
ALLOW	ANY	TCP	ANY	80
ALLOW	ANY	TCP	ANY	25
ALLOW	ANY	TCP	ANY	110
DENY	ANY	UDP	ANY	23
DENY	ANY	TCP	ANY	3389

Packet Filtering FW Rules

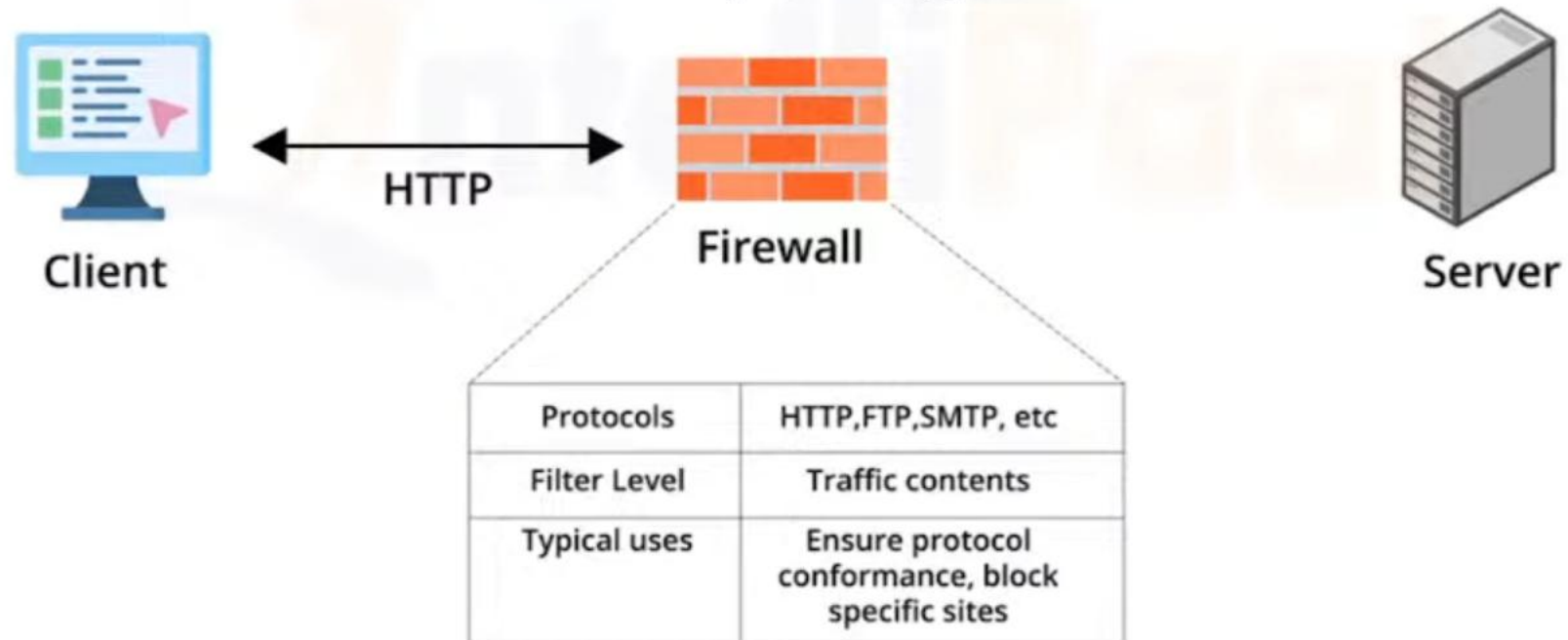
- Example 1: block incoming and outgoing datagrams with IP protocol field = 17 and with either source or dest port = 23.
 - All incoming and outgoing UDP flows and telnet connections are blocked.
- Example 2: Block inbound TCP segments with ACK=0.
 - Prevents external clients from making TCP connections with internal clients, but allows internal clients to connect to outside.

Firewalls - layer 3

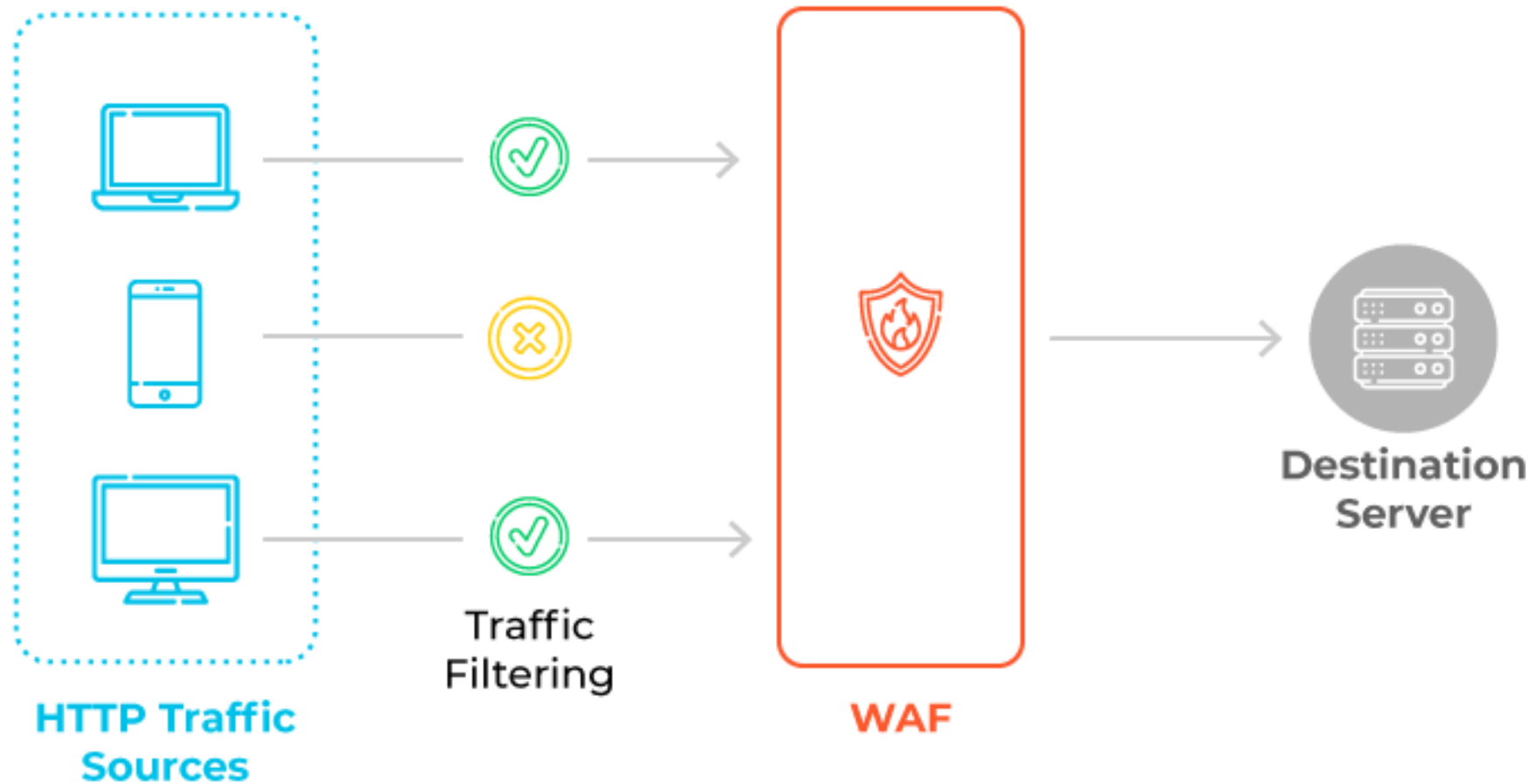


Application gateways (proxy FW)

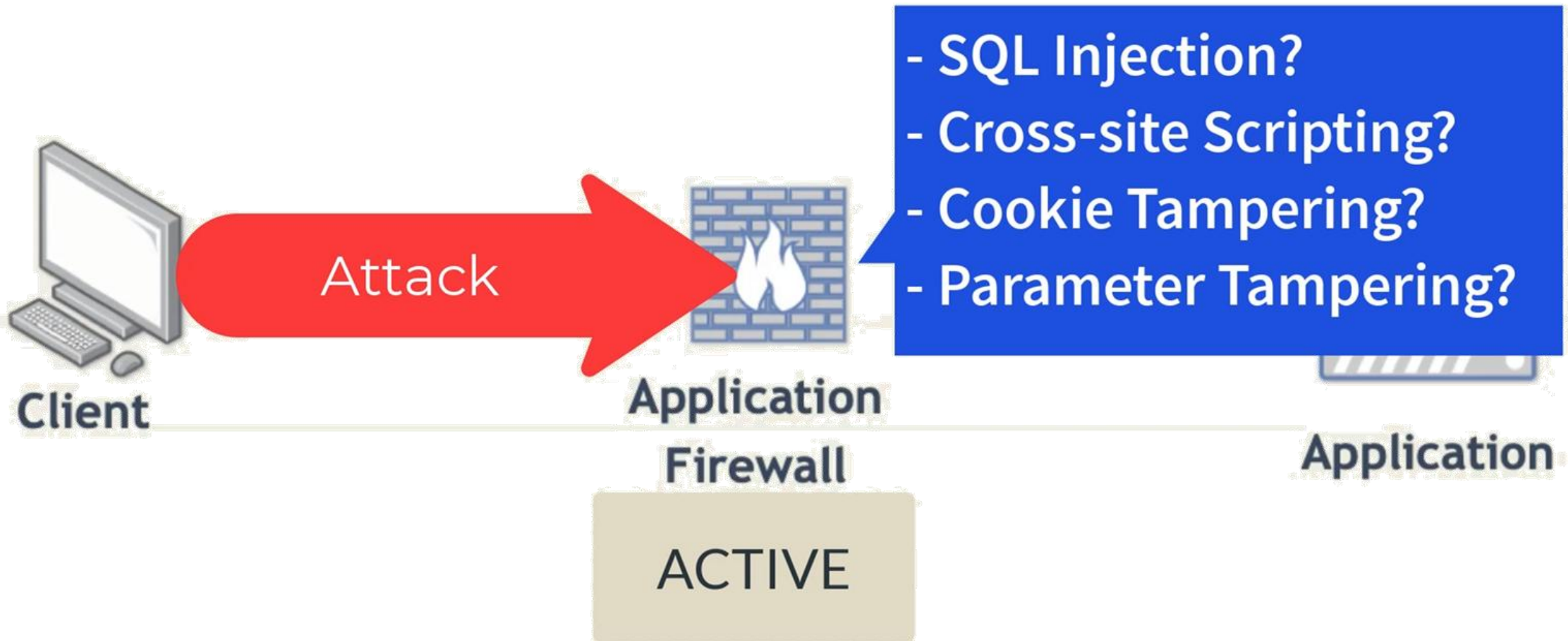
A proxy firewall is an early type of firewall device that acts as a gateway from one network to another for a specific application.



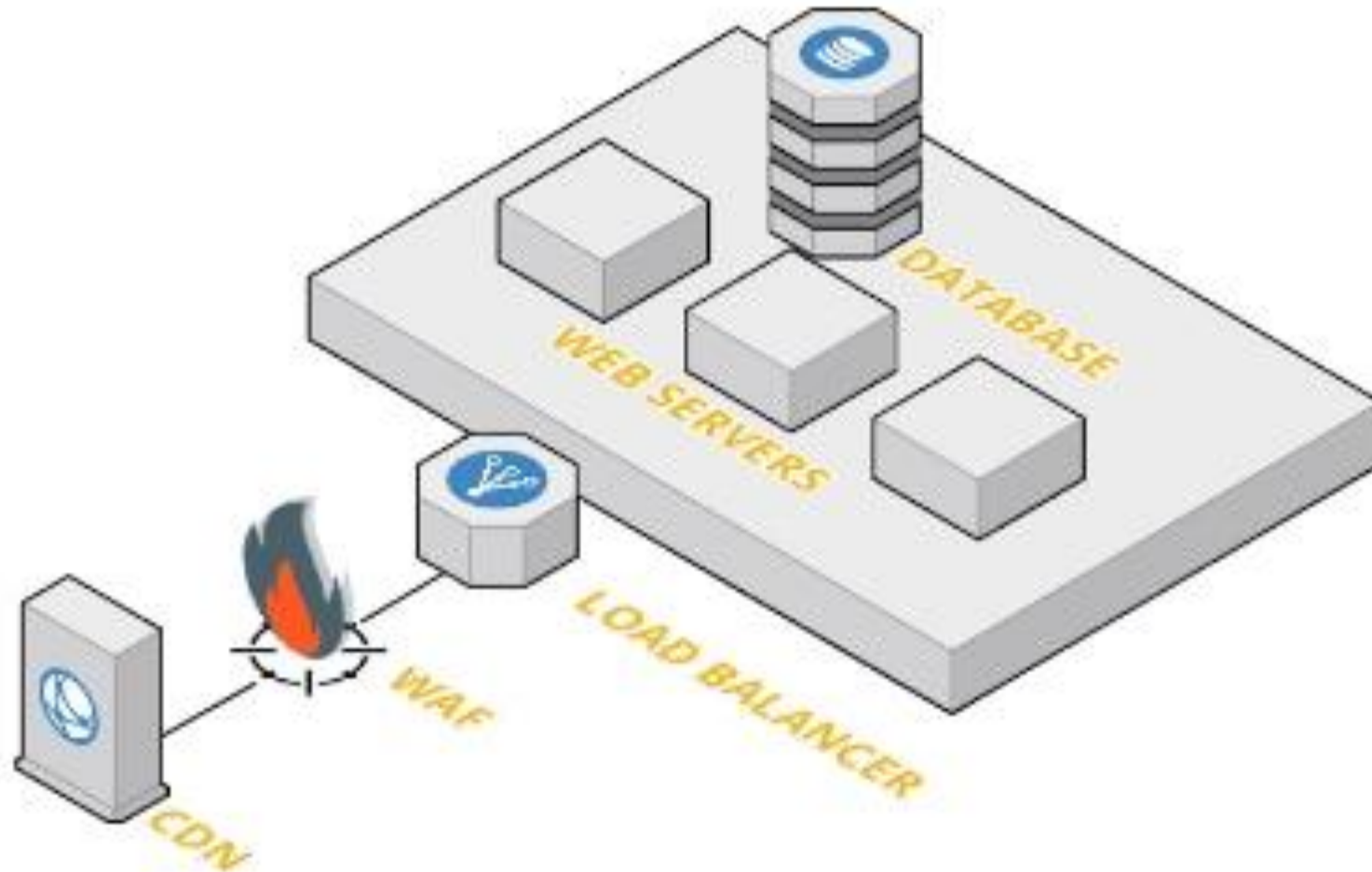
Web Application Firewalls (WAF)



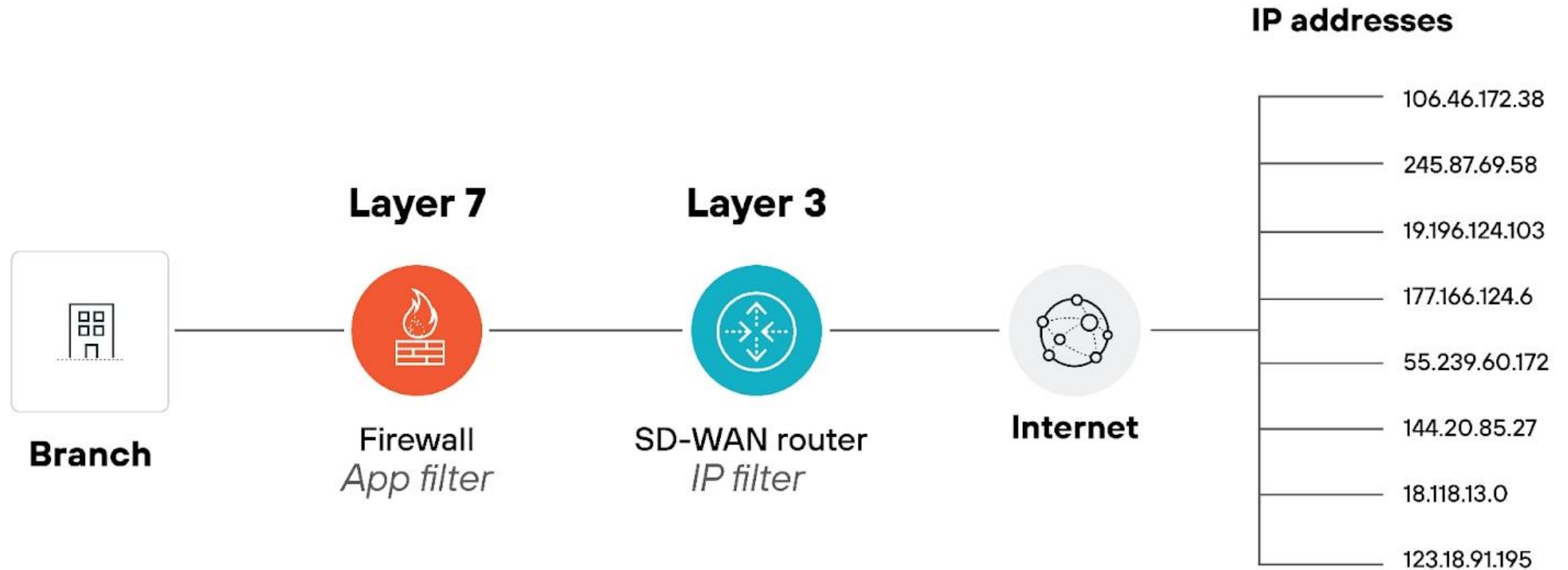
Web Application Firewalls (WAF)



Web Application Firewalls (WAF)



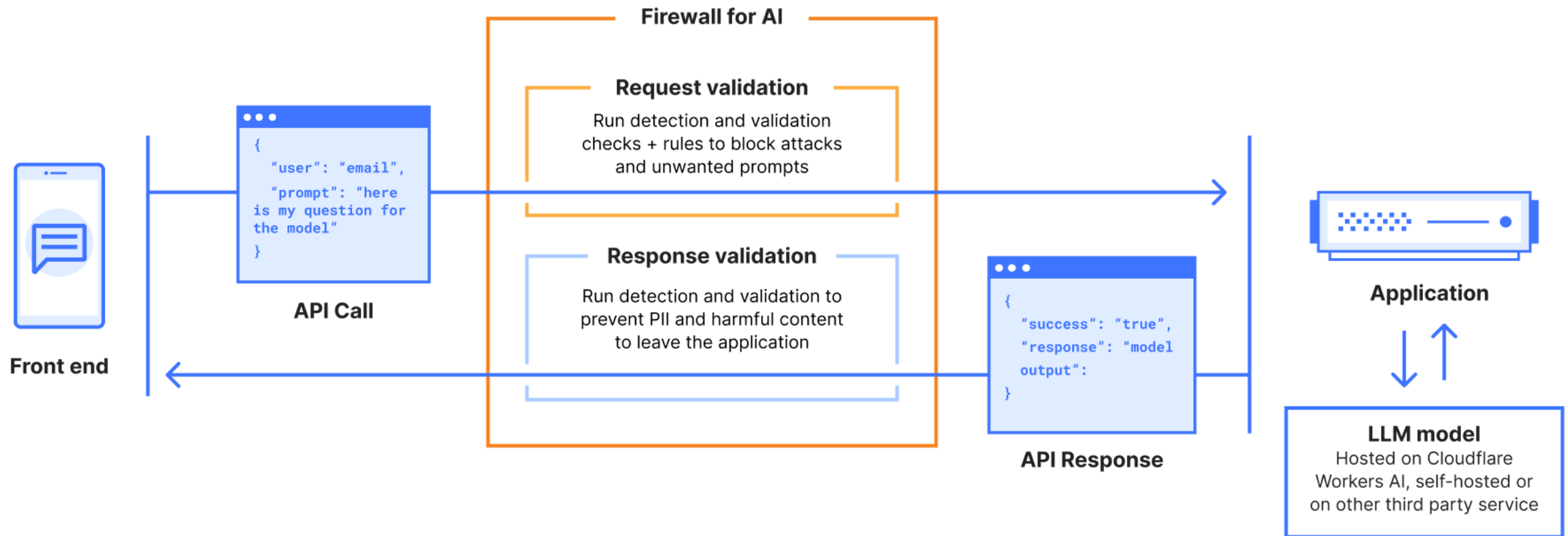
FW L3 + L7



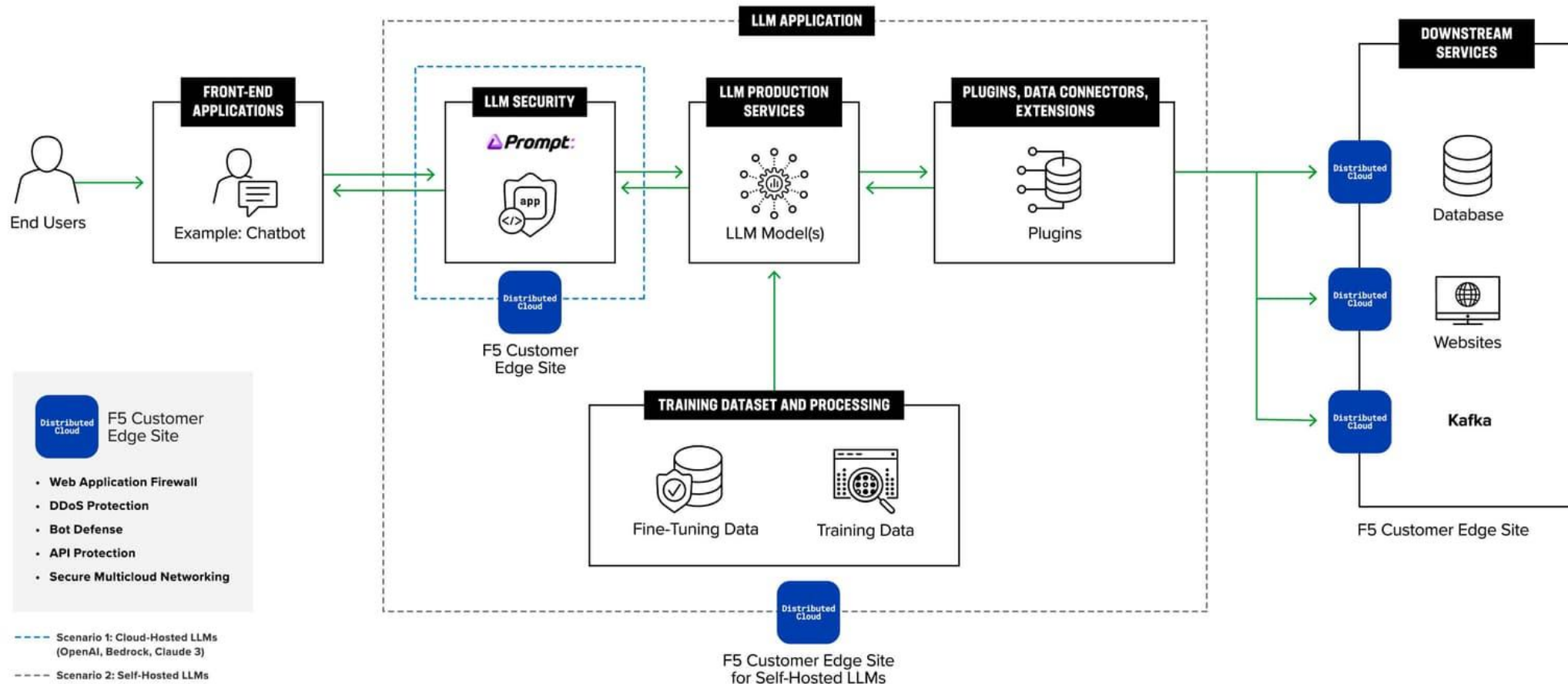
FW L3 + L7

	Layer 3 firewalls	Layer 7 firewalls
Traffic filtering	Based on IP addresses and ports	Examines contents within data packets
Performance	High throughput; handles large volumes of traffic	Lower throughput due to deep packet inspection
Speed of inspection	Fast; simple filtering based on addresses & ports	Slower; detailed analysis of packet contents
Security focus	General threats; basic access control	Complex, application-targeted threats
Use case	Suitable where speed and volume are critical	Needed for sensitive applications needing deep inspection
Typical threat mitigation	Blocks threats based on source/destination info	Mitigates threats exploiting application vulnerabilities like SQL injections, XSS

GEN AI FW



GEN AI FW

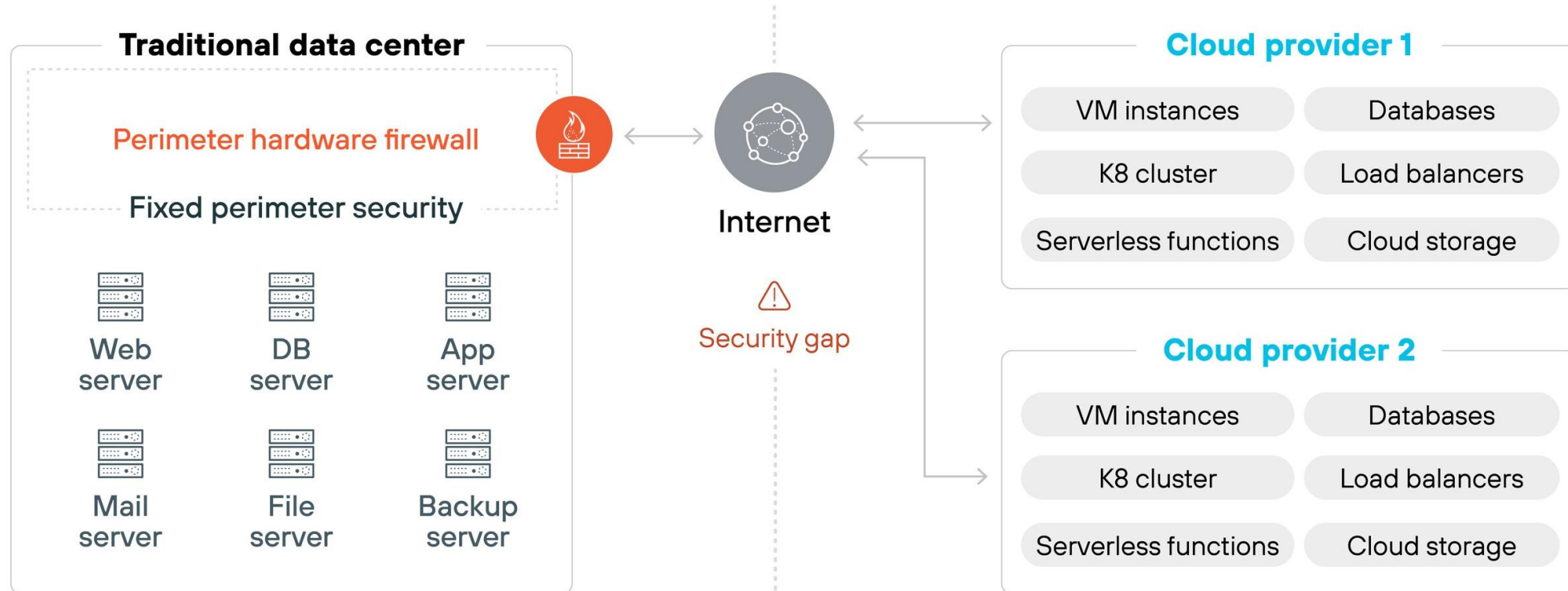


Prompt Security firewall for AI sits in the path of inbound inquiries to generative AI LLMs and outbound responses to stop inbound attacks and outbound data exfiltration.

AI-FW vs NG-FW

Capability	AI Firewall	Next-Generation Firewall (NGFW)
Focus Area	LLMs, AI applications, GenAI workflows	Network layer, application protocols, threat blocking
Content Inspection	Natural language inputs/outputs, AI-specific logic	Packets, ports, known malware, protocol headers
Key Features	Prompt filtering, output redaction, API control	IDS/IPS, antivirus, WAF, URL filtering
Deployment Layer	App/API/model layer	Network perimeter, routers, VPN endpoints
Threat Types	Prompt injection, data leaks, model abuse	DDoS, malware, phishing, ransomware
Target Users	AI/ML developers, security architects, SOC teams	Network admins, infrastructure and IT security teams
Example Use Case	Protecting a GenAI-powered customer support chatbot	Blocking phishing and malware from a remote office

Firewalls across environments



Traditional model characteristics

- Fixed perimeter design
- Visibility centered on on-premises environments
- Policy enforcement at centralized boundaries

Cloud characteristics:

- Workloads span multiple cloud providers
- Dynamic scaling and auto-provisioning
- Serverless and ephemeral workloads



אבטחת מידע ממוחשב